Summer Research project

Day to Day progress:

1. **Static Contact angle measurement**
- Learned to measure the static contact angle of a droplet using a goniometer.
- Understood the significance of contact angle in characterizing surface wettability
2. **Dynamic and Hysteresis Contact Angle Measurement**
- Learned to measure advancing and receding contact angles.
- Practiced calculating contact angle hysteresis using the tilting (inclined plane) method.
3. **Surface Defect Analysis:**
- Practiced measuring dynamic contact angles on surfaces with and without defects.
- Understood how defects influence contact angle and droplet behavior
4. **Surface Modification:**
- Converted a superhydrophilic surface into a superhydrophobic one.
- Used silica particles spraying method to achieve superhydrophobicity.
5. **Research paper reading material:**
- Surface-wetting characterization using contact-angle measurements.

Static Contact angle measurement

Sample Preparation Protocol:

Before performing contact angle measurements, it is essential to ensure the surface is clean to obtain accurate and reproducible results. The following cleaning procedure was followed:

1. The sample surface was cleaned sequentially using acetone, isopropyl alcohol (IPA), and deionized (DI) water to remove any organic and particulate contaminants.
2. After rinsing, the sample was dried using a nitrogen gas jet.
3. For the measurement, the droplet used was deionized water, ensuring high purity for consistent results.

<u>Sessile-drop goniometry:</u>

- The method of measuring contact angles in the protocol is called sessile-drop goniometry. It is performed by recording a video of a water drop on a solid surface and determining the contact angle from the images of the video by a fitting procedure. The following measurement is performed on goniometer. Sessile-drop goniometry.
- A drop of water on an ideal solid substrate. Young contact angle (θ_{Young}) is determined by a balance of the horizontal projection of the surface tension of the water along the solid surface ($\gamma \cos\theta_{Young}$) and interfacial tensions γ_{sv} and γ_{sl}.

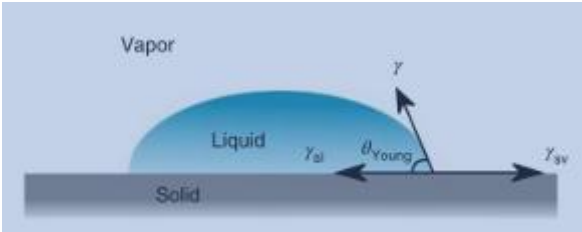
Goniometer

Dynamic and Hysteresis Contact Angle Measurement:

1. **Different wetting characterization methods:**

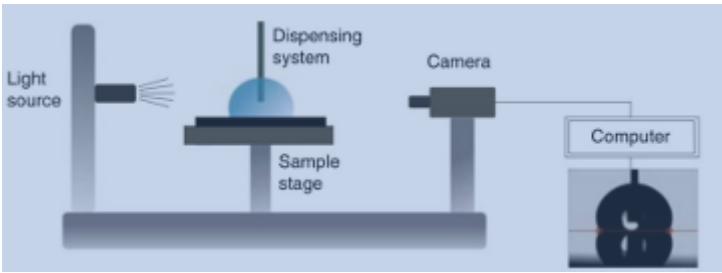
- **Sessile-drop goniometry :**

The volume of a drop deposited on the measured surface is increased, and the value of the ACA is obtained from the advancing contact line. For the RCA, the volume of the drop is reduced, and the value of the RCA is obtained from the receding contact line.



- **Tilting plate :**

Contact angles are measured from the leading edge and the trailing edge of a distorted drop on an inclined plane when the drop starts sliding. The tilt angle at which the drop starts sliding is called 'sliding angle', and is a measure of droplet mobility, the side with moved by tilting measures ACA and the other one is for RCA.



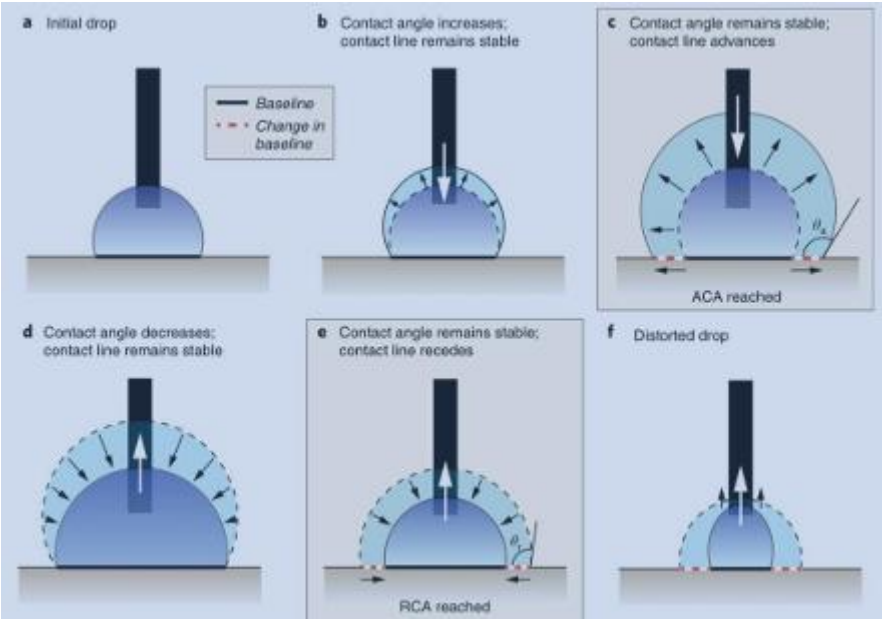
2. **Hysteresis Angle:**

- The difference between our ACA and RCA is equal to the inclined angle with which we have tilted the plane/our sample with, and is called hysteresis angle.

3. **<u>Apparent and intrinsic contact angle:</u>**

- **Apparent Contact angle :**

This is the contact angle you observe or measure on a real surface, which may be rough or chemically heterogeneous. It can be influenced by surface roughness, contamination, or droplet size.



- **Intrinsic Contact angle :**

This is the ideal contact angle defined for a perfectly smooth, flat, and chemically homogeneous surface. It comes from Young's equation:

$$\cos\theta_{\text{Young}} = \frac{\gamma_{\text{sv}} - \gamma_{\text{sl}}}{\gamma},$$

<u>SURFACE DEFECT ANALYSIS:</u>

1. **<u>Dynamic contact angles:</u>**

- I practiced measuring dynamic contact angles on anonymous surfaces first and then on a previous superhydrophobic surface, with or without defect, here is the observations i got, I calculated average ACA and RCA for the same, and their standard deviation.

Defect presence	Advanced contact angle				Receding contact angle			
with defect	Left	Right	Avg ACA		Left	Right	Avg RCA	
1_1	169.992	169.046	169.519	AVG ACA	168.742	168.0435	168.3928	AVG RCA
1_1b	168.366	162.561	165.4635	167.0391	160.747	159.193	159.97	163.4755
1_1c	171.909	166.759	169.334	STD	168.294	165.79	167.042	STD
2_1	164.302	166.329	165.3155	1.712359	154.022	162.378	158.2	6.041596
2_1-1					169.334	167.179	168.2565	
without defect	Left	Right	Avg ACA		Left	Right	Avg RCA	
1	166.293	171.781	169.037	AVG ACA	163.434	165.684	164.559	AVG RCA
2	166.293	170.86	168.5765	168.9908	161.781	159.274	160.5275	166.4836
3	172.682	167.417	170.0495	STD	172.504	166.211	164.9425	STD
4	169.3575	171.367	170.3623	0.66555	165.233	167.453	168.4053	1.519098

2. **<u>Contact angle impact:</u>**

- Surface defects such as roughness, scratches, or chemical heterogeneities affect how a droplet interacts with the surface. On defective surfaces, the contact line may get pinned, causing irregular advancing or receding motion. This results in increased contact angle hysteresis and less predictable droplet shapes. During experiments, I observed that defects can locally alter the contact angle, leading to asymmetric droplet profiles and affecting dynamic measurements.

SURFACE MODIFICATION:

Step 1 Converting Hydrophilic Surface to Superhydrophilic:

Initially, we worked with a natural glass surface, which is typically hydrophilic in nature. To enhance its wettability and convert it into

a superhydrophilic surface, we used oxygen plasma treatment. The plasma process increases surface energy by introducing polar functional groups and removing contaminants, enabling water droplets to spread instantly upon contact. This step created a high energy surface that served as the base for further modifications.

Step 2 Coating with Mirror Coating Zero and Spraying Alcohol-Silica Mixture:

To convert the superhydrophilic surface into a superhydrophobic one, we first applied a thin layer of mirror coating zero, which acts as a base layer to assist in surface roughness enhancement and chemical compatibility. Following this, we sprayed a mixture of alcohol and silica nanoparticles using a jet spray method. The silica particles introduced micro and nano-scale roughness, while the alcohol served as a carrier for uniform dispersion.

Step 3 Achieving Superhydrophobicity:

The combination of surface roughness from the silica particles and the low surface energy layer from mirror coating zero enabled the transition from superhydrophilic to superhydrophobic. The treated surface exhibited water contact angles greater than 150°, with water droplets rolling off easily, indicating very low adhesion.

Observations of superhydrophobic surface sample 1:

	Advanced contact angle				Receding contact angle			
	Left	Right	Avg		Left	Right	Avg	
Sample 1	171.101	166.053	168.577	final avg	171.265	164.641	167.953	final avg
Sample 2	173.884	168.511	171.1975	170.3761	170.603	167.441	169.022	167.9871
Sample 3	169.066	172.454	170.76	STD	168.12	167.829	167.9745	STD
Sample 4	172.189	169.751	170.97	1.050185	166.866	167.132	166.999	0.715661

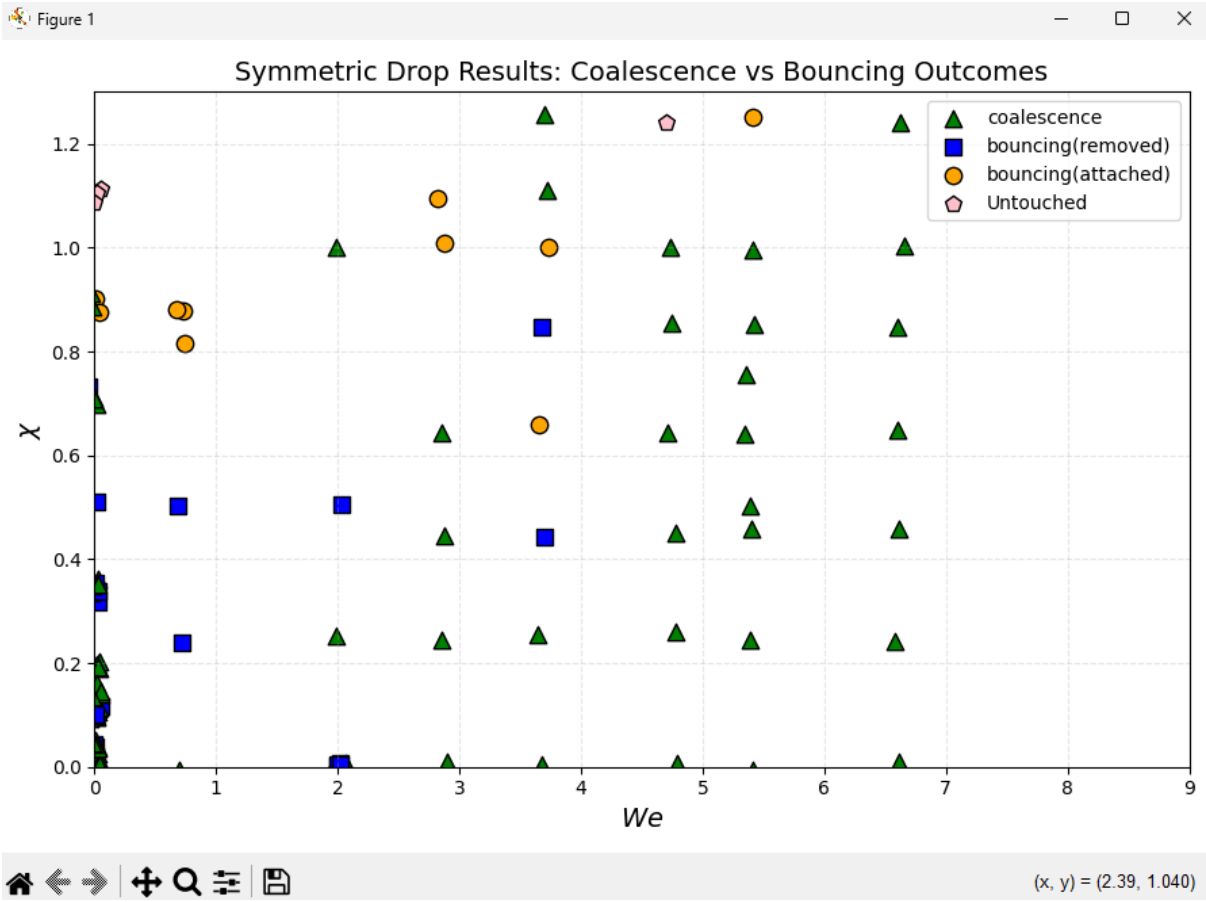
- We observed the contact angle for the super hydrophobic surface we made before we made defects on its surface on different parts of its surface.

The following results focused on acquiring foundational knowledge related to contact angle measurement, surface wettability, and surface modification techniques. I learned how to accurately measure both static and dynamic contact angles using a goniometer, including advancing, receding, and hysteresis angles. I also explored how surface properties—such as hydrophilic, hydrophobic, superhydrophilic, and superhydrophobic behaviors—affect droplet behavior.

In addition, I gained hands-on experience using relevant software tools to analyze droplet shape and measure angles precisely. The week also introduced me to surface treatment techniques such as oxygen plasma coating for enhancing hydrophilicity, and the use of silica nanoparticles and mirror coating zero to achieve superhydrophobicity.

Weber no. (we)	0.7201	2.012	2.87	3.6856	4.746	5.396	6.616
χ =0	Coalesce (8) Bouncing (3)	Bouncing (4) Coalesce (6)	Bouncing (removed) Coalesce (2)	Coalesce	Coalesce(3)	Coalesce(2)	Coalesce
χ = 0.25	Bouncing (9) Coalesce (6)	Bouncing (1) Coalesce (3)	Coalesce	Coalesce	Coalesce	Coalesce	Coalesce
χ = 0.45			Coalesce	Bouncing (removed)	Coalesce	Coalesce	Coalesce
χ = 0.50	Bouncing (6)	Bouncing (4) Coalesce (1)					
χ = 0.65			Coalesce	Bouncing (attached)	Coalesce	Coalesce	Coalesce
χ =0.75							
χ =0.85				Bouncing (removed)	Coalesce	Coalesce	Coalesce
χ =1	Bouncing (3)	Coalesce (4)	Coalesce Bouncing (attached)2	Bouncing (attached)	Coalesce	Coalesce	Coalesce
χ = 1.1			Bouncing (attached)	Coalesce			
χ = 1.25				Coalesce	Coalesce Untouched	Bouncing (attached)2	Coalesce

Symmetric Drop Results



Data Representation Using Metplotlib